Final Project (35 Points)

Due: Sunday 10th May, 11:59 pm

The final project is the highlight of this course. Since software development is usually a team effort, you can work with **three or four classmates** on this project. Everyone in the group is expected to contribute equally to the design and implementation.

Idea: Creating a Business Database from Scratch

Project overview: A business owner approached you for help in creating a new database from scratch to streamline their company's operations. The existing business data is stored in an Excel file, which can be downloaded from Canvas (CompanyData.xlsv). The dataset contains 31 columns and 1,000 rows. You need to design a robust and customized database solution tailored to their specific needs. For this project, you will use MySQL (MySQL Workbench) as the Database Management System (DBMS).

Final output: You must submit a complete **report** that clearly documents all phases of the project via **Canvas**. Please follow the guidelines below:

a) Report Requirements

1. **Explanation of Each Phase**: Provide a detailed description of what was done in each phase of the project.
2. **Screenshots**: Include a screenshot of the relevant output for **each phase**.
3. **Team Contribution Table**: Add a **table** in your report that clearly shows **which parts** of the project were completed by **you** and **your teammate**.

   Note: You will be **graded based on this table**, so make sure it's clear and accurate.

b) Files to Upload (Alongside the Report)

1. **SQL Output Files**: Save the output of each phase in **.sql format**.
2. **Exported Code Outputs**: Export any output generated by your code (e.g., query results) as **separate files**. Use **.csv format**, if applicable.
3. **File Naming**: Name each exported file **clearly and appropriately**. Example file names: db_view.sql for SQL script file, db_view.csv for exported result of the query.
4. **info.txt File**: Create an info.txt file that contains clear instructions on how to run your project from scratch. This should include: software requirements (e.g., MySQL version), setup steps (e.g., how to import the database), how to run the queries or scripts and any dependencies or configurations needed.

Hint and specifications: After analyzing the data structure and identifying the necessary actions to assist the business, you should have outlined the process into the following steps:

- **Phase1: Database Creation & Data Loading (5 points):** Before creating the database, the data in the Excel file should be thoroughly reviewed, cleaned, and then saved in CSV format. Then, the next step is to create a database and import the data into it. This can be done in two ways:

    1. Using the Table Import Wizard in MySQL Workbench (or another SQL tool with a similar import feature) to import the data.
    2. Using the LOAD DATA INFILE command in MySQL Shell to load the data into a table.

    To use the second method, start by cleaning the data in the Excel file and saving it as a CSV file. The cleaning process includes:

    1. Ensuring date formats are in "yyyy-mm-dd."
    2. Verifying that the correct number format is used, with no currency symbols, commas, or other symbols.

- **Phase2: Database Diagram Design (ER) (5 points):** The next step is to create an ER (Entity-Relationship) Diagram. ER diagrams utilize symbols to represent entities, attributes, and relationships, effectively visualizing how the entities in the database are interconnected. You can create your entity relationship diagram any way you'd like but allow me to suggest a few!
    - If you're the pencil-and-paper type, you can draw your diagram, take a picture, and upload it as a separate file.
    - If you're digitally inclined, you can use any software that helps you draw the types of shapes you'd like to draw to create and export diagrams.
    - Or you can use Reverse Engineer feature of MySQL to create the ER Diagram.

- **Phase3: Normalization and Denormalization (5 points):** After examining the imported data, you'll notice that it is denormalized. This leads to the next step: Data Normalization. Normalization involves organizing the data according to specific attributes within a larger data model. The primary goals of database normalization are to remove redundant data, reduce the risk of data modification errors, and simplify the querying process. The SQL technique you can use to normalize your data is the CTAS (CREATE TABLE AS SELECT) method.

- **Phase4: Table Alterations (5 points):** After creating the ER Diagram and identifying the required constraints (such as primary keys, foreign keys, and attribute rules), you must modify your existing tables to match the structure defined in your design.

  In this phase, you will use ALTER TABLE statements to update your tables. The goal is to ensure that your database enforces data integrity and correctly represents the relationships between entities. Your work in this phase should include:

  - Adding Primary Keys to uniquely identify each record in a table.
  - Adding Foreign Keys to establish relationships between tables based on your ER diagram.
  - Applying constraints such as NOT NULL and UNIQUE where appropriate.
  - Modifying data types if necessary to ensure consistency (e.g., dates, numeric values).
  - Renaming or removing columns if they do not align with the normalized design.

  You should provide the SQL commands used for each modification and briefly explain how these changes reflect your ER diagram.

- **Phase5: Views, Triggers, and Stored Procedures (5 points):** At this stage, the database should be fully structured and aligned with the business requirements. The business owner has requested the implementation of the following features:

  - **View:** Create a view that displays the total quantity sold and total revenue generated by each employee. The view should combine relevant tables (e.g., orders, order details, employees) and use appropriate aggregate functions.

  - **Trigger:** Implement a trigger on the products table that automatically updates (reduces) the stock quantity whenever a new order is inserted. Clearly specify when the trigger is activated (e.g., AFTER INSERT).

  - **Stored Procedure:** Develop a stored procedure that identifies products with low stock (below a specified threshold) and returns a list of items that need to be restocked.

- **Phase6: User Management and Privileges (5 points):** The business owner has requested the creation of two users—a Database Administrator (DBA) and a Data Analyst—with specific access privileges to the database. You are required to create two users:

  - A DBA user (e.g., "AlexSmith") with full administrative privileges on the database.
  - An Analyst user (e.g., "JamieLee") with read-only access.
- Assign appropriate permissions using SQL commands such as GRANT and REVOKE:

  - The DBA should have full privileges (e.g., ALL PRIVILEGES) on the database.
  - The Analyst should only have SELECT permissions on relevant tables.

    - Ensure that privileges are applied correctly and follow best practices for database security.
- This phase emphasizes the importance of controlling access to the database as part of overall database design and security.
- **Phase7: Database Backup (5 points):** The main goal of a database backup is to create a complete copy of the database (data and structure) at a specific point in time. This ensures that the database can be restored in case of data loss, corruption, or system failure. Backups can be stored in different secure locations, such as:
    - On-premises servers
    - External hard drives or USB drives
    - Network-attached storage (NAS)
    - Cloud storage services (e.g., Amazon S3, Google Cloud Storage, Microsoft Azure Blob Storage)
    - and so many other locations.

  In this project, you should perform a backup using MySQL's **Data Export feature** (available in MySQL Workbench or local instance tools) and save the backup file to a secure location. You must submit evidence of the backup process (e.g., exported file or screenshots) as part of your work.

Well done! You are now ready to design, build, and manage real-world databases with confidence.